Lesson 11: Limit Theorems

AI/ML Mathematics

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1 Motivation and Overview

The limit theorems of probability theory are the workhorses behind nearly every convergence guarantee in reinforcement learning. The law of large numbers tells us that sample averages converge to expectations; the central limit theorem quantifies the rate of this convergence; and the monotone and dominated convergence theorems justify the interchange of limits and expectations that appears implicitly in almost every derivation involving infinite sums of discounted rewards.

This lesson proves six fundamental results:

1. The **Weak Law of Large Numbers** (WLLN),
2. The **Strong Law of Large Numbers** (SLLN, Kolmogorov),
3. The **Central Limit Theorem** (CLT),
4. The **Monotone Convergence Theorem** (MCT),
5. **Fatou's Lemma**,
6. The **Dominated Convergence Theorem** (DCT, Lebesgue).

Why This Matters for RL

- **WLLN**: justifies Monte Carlo estimation—averaging n independent returns from a state converges in probability to $V^\pi(s)$.
- **SLLN**: guarantees the almost sure convergence required by stochastic approximation theory (Robbins–Monro), underpinning TD and Q-learning proofs.
- **CLT**: provides $O(1/\sqrt{n})$ error rates and confidence intervals for policy evaluation via Monte Carlo.
- **DCT**: justifies writing $V^\pi(s) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid S_0 = s]$ as $\sum_{t=0}^{\infty} \gamma^t \mathbb{E}[R_{t+1} \mid S_0 = s]$, and underpins the interchange of limit and expectation in value function convergence arguments.

Prerequisites. Lesson 5 (expectation, linearity), Lesson 7 (variance, independence), Lesson 9 (Markov, Chebyshev, Jensen inequalities), Lesson 10 (modes of convergence, Borel–Cantelli).

2 References

The following sources complement this lesson:

- Durrett [2019], Chapter 2 (Sections 2.2–2.4): laws of large numbers and the CLT with proofs via characteristic functions.
- Billingsley [1995], Sections 21–22 (MCT, Fatou, DCT) and Sections 27–28 (CLT via Lindeberg–Feller).
- Williams [1991], Chapter 5: a martingale-based approach to the SLLN.
- Folland [1999], Section 2.4: a clean treatment of the three convergence theorems for integration.
- Shiryaev [2016], Chapter III, Sections 3–4: laws of large numbers and CLT with applications.

3 Convergence Theorems for Integration

We begin with the three fundamental theorems that govern the interchange of limits and expectations (Lebesgue integrals). These are purely analytic results that hold in any measure space; we state them in the probability setting.

3.1 Monotone Convergence Theorem

Monotone Convergence Theorem

Theorem 3.1 (Monotone Convergence Theorem). *If $0 \le X_1 \le X_2 \le \cdots$ a.s. and $X_n \xrightarrow{a.s.} X$, then*

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X]. \tag{1}$$

Both sides may equal $+\infty$.

Proof. **Step 1: Upper bound.** Since $X_n \le X$ a.s. for every n, monotonicity of expectation gives $\mathbb{E}[X_n] \le \mathbb{E}[X]$. The sequence $\{\mathbb{E}[X_n]\}$ is non-decreasing (since $X_n \le X_{n+1}$), so $L := \lim_n \mathbb{E}[X_n]$ exists in $[0,\infty]$ and $L \le \mathbb{E}[X]$.

Step 2: Lower bound via simple functions. Let $s = \sum_{k=1}^m c_k \mathbf{1}_{A_k}$ be any non-negative simple function with $0 \le s \le X$ a.s. Fix $\alpha \in (0,1)$ and define $B_n = \{X_n \ge \alpha s\}$. Since $X_n \uparrow X \ge s > \alpha s$ on $\{s > 0\}$, the sets B_n increase to a set of full measure (up to the null set where $s = 0$ and $X = 0$). Then

$$\mathbb{E}[X_n] \ge \mathbb{E}[X_n \mathbf{1}_{B_n}] \ge \alpha\, \mathbb{E}[s\, \mathbf{1}_{B_n}]. \tag{2}$$

As $n \to \infty$, continuity of measure from below gives $\mathbb{P}(B_n) \to 1$, and hence $\mathbb{E}[s\, \mathbf{1}_{B_n}] \to \mathbb{E}[s]$ (for each block A_k, $\mathbb{P}(A_k \cap B_n) \to \mathbb{P}(A_k)$). So $L \ge \alpha\, \mathbb{E}[s]$.

Letting $\alpha \uparrow 1$ gives $L \ge \mathbb{E}[s]$. Taking the supremum over all simple $s \le X$ yields $L \ge \mathbb{E}[X]$ (since $\mathbb{E}[X]$ is defined as this supremum by the Lebesgue construction).

Conclusion. $L \le \mathbb{E}[X]$ and $L \ge \mathbb{E}[X]$, so $L = \mathbb{E}[X]$. □

Remark 3.2 (Non-Negativity is Essential). The theorem fails without the non-negativity assumption. For example, let $X_n = -\mathbf{1}_{(n,\infty)}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ with Lebesgue measure. Then $X_n \uparrow 0$ pointwise, but each $\mathbb{E}[X_n]$ is undefined (or $-\infty$), so the conclusion does not hold in the standard sense. In the probability setting with $\mathbb{P}(\Omega) = 1$, the issue arises when $\mathbb{E}[X_1] = -\infty$.

3.2 Fatou's Lemma

Fatou's Lemma

Theorem 3.3 (Fatou's Lemma). *If $X_n \ge 0$ a.s. for all n, then*

$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n\Big] \le \liminf_{n\to\infty} \mathbb{E}[X_n]. \tag{3}$$

Proof. Define $Y_m = \inf_{n\ge m} X_n$ for each $m \ge 1$. Then:

1. $0 \le Y_1 \le Y_2 \le \cdots$ (the infimum is taken over a smaller set as m increases).
2. $Y_m \uparrow \liminf_n X_n$ a.s. (by the definition of lim inf).
3. $Y_m \le X_m$ a.s. (since the infimum is at most any term in the sequence).

By the Monotone Convergence Theorem (Theorem 3.1):

$$\mathbb{E}[\liminf_n X_n] = \mathbb{E}[\lim_m Y_m] = \lim_m \mathbb{E}[Y_m]. \tag{4}$$

Since $Y_m \le X_m$ a.s., we have $\mathbb{E}[Y_m] \le \mathbb{E}[X_m]$, and therefore

$$\lim_m \mathbb{E}[Y_m] \le \liminf_m \mathbb{E}[X_m] = \liminf_n \mathbb{E}[X_n]. \tag{5}$$

□

Remark 3.4 (Strict Inequality is Possible). Let $X_n = n\, \mathbf{1}_{(0,1/n)}$ on $[0,1]$ with Lebesgue measure. Then $X_n \to 0$ a.s., so $\liminf X_n = 0$ and $\mathbb{E}[\liminf X_n] = 0$. But $\mathbb{E}[X_n] = 1$ for all n, so $\liminf \mathbb{E}[X_n] = 1 > 0$. Fatou's lemma gives $0 \le 1$, which is correct but not an equality.

3.3 Dominated Convergence Theorem

Dominated Convergence Theorem

Theorem 3.5 (Dominated Convergence Theorem (Lebesgue)). *If $X_n \xrightarrow{a.s.} X$ and there exists an integrable random variable $Y \geq 0$ with $\mathbb{E}[Y] < \infty$ such that $|X_n| \leq Y$ a.s. for all n, then*

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X] \quad \textit{and} \quad \mathbb{E}[|X_n - X|] \to 0. \tag{6}$$

Proof. Since $|X_n| \leq Y$ a.s. and $X_n \to X$ a.s., we have $|X| \leq Y$ a.s., so X is integrable.

Consider the non-negative sequences $Y + X_n \geq 0$ and $Y - X_n \geq 0$. Apply Fatou's lemma (Theorem 3.3) to each:

$$\mathbb{E}[Y + X] = \mathbb{E}[\liminf_n (Y + X_n)] \leq \liminf_n \mathbb{E}[Y + X_n] = \mathbb{E}[Y] + \liminf_n \mathbb{E}[X_n], \tag{7}$$

$$\mathbb{E}[Y - X] = \mathbb{E}[\liminf_n (Y - X_n)] \leq \liminf_n \mathbb{E}[Y - X_n] = \mathbb{E}[Y] - \limsup_n \mathbb{E}[X_n]. \tag{8}$$

From (7): $\mathbb{E}[X] \leq \liminf_n \mathbb{E}[X_n]$. From (8): $\mathbb{E}[X] \geq \limsup_n \mathbb{E}[X_n]$. Together: $\lim_n \mathbb{E}[X_n] = \mathbb{E}[X]$.

For L^1 convergence: apply the above argument to the sequence $|X_n - X|$, which converges a.s. to 0 and is dominated by $2Y$ (since $|X_n - X| \leq |X_n| + |X| \leq 2Y$). This gives $\mathbb{E}[|X_n - X|] \to 0$. □

Example 3.6 (Interchanging Sum and Expectation). Let $R_1, R_2, \ldots$ be bounded rewards with $|R_t| \leq R_{\max}$, and let $\gamma \in [0, 1)$. Define $G_n = \sum_{t=0}^{n-1} \gamma^t R_{t+1}$. Then $G_n \to G := \sum_{t=0}^{\infty} \gamma^t R_{t+1}$ pointwise, and $|G_n| \leq R_{\max} \sum_{t=0}^{n-1} \gamma^t \leq R_{\max}/(1-\gamma)$. Taking $Y = R_{\max}/(1-\gamma)$, the DCT gives

$$\mathbb{E}[G] = \lim_n \mathbb{E}[G_n] = \lim_n \sum_{t=0}^{n-1} \gamma^t \mathbb{E}[R_{t+1}] = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}[R_{t+1}]. \tag{9}$$

This justifies the interchange of infinite sum and expectation used in the definition of $V^\pi(s)$.

DCT in Value Function Theory

The Dominated Convergence Theorem is used repeatedly in RL theory:

- **Defining V^π:** The value function $V^\pi(s) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid S_0 = s]$ is well-defined because partial sums are dominated by $R_{\max}/(1-\gamma)$.
- **Bellman equations:** Deriving $V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a)[R(s,a,s') + \gamma V^\pi(s')]$ requires exchanging expectation and limit.
- **Convergence of operators:** When proving that the Bellman operator preserves continuity or that iterates converge, the DCT justifies passing limits through conditional expectations.

4 Laws of Large Numbers

4.1 Weak Law of Large Numbers

Weak Law of Large Numbers

Theorem 4.1 (Weak Law of Large Numbers). *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then*

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu. \tag{10}$$

Proof. By linearity of expectation, $\mathbb{E}[\bar{X}_n] = \mu$. By independence and identical distributions (Lesson 7):

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}. \tag{11}$$

By Chebyshev's inequality (Lesson 9, Theorem 2.3):

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0 \quad \text{as } n \to \infty, \tag{12}$$

for any fixed $\varepsilon > 0$. □

WLLN and Monte Carlo Estimation

The WLLN is the theoretical foundation of Monte Carlo policy evaluation. Given n independent episodes starting from state s, the sample mean of observed returns $\bar{G}_n(s) = \frac{1}{n}\sum_{i=1}^{n} G_i(s)$ satisfies $\bar{G}_n(s) \xrightarrow{\mathbb{P}} V^\pi(s)$. This guarantees that the every-visit Monte Carlo estimate converges in probability, requiring only a finite variance assumption on the return distribution.

4.2 Strong Law of Large Numbers

Strong Law of Large Numbers

Theorem 4.2 (Strong Law of Large Numbers (Kolmogorov)). *Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$. Then*

$$\bar{X}_n \xrightarrow{a.s.} \mu. \tag{13}$$

Proof. We prove the result under the stronger assumption $\mathbb{E}[X_1^4] < \infty$; the general case (requiring only $\mathbb{E}[|X_1|] < \infty$) uses a truncation argument (see Remark 4.3).

Let $Y_i = X_i - \mu$, so $\mathbb{E}[Y_i] = 0$, $\mathbb{E}[Y_i^2] = \sigma^2$, and $S_n = \sum_{i=1}^{n} Y_i = n(\bar{X}_n - \mu)$.

Step 1: Fourth moment bound. Expand $S_n^4 = (\sum_{i=1}^{n} Y_i)^4$. The expectation $\mathbb{E}[S_n^4]$ receives contributions from:

- Terms $\mathbb{E}[Y_i^4]$: there are n terms, each equal to $\mathbb{E}[Y_1^4]$.
- Terms $\mathbb{E}[Y_i^2 Y_j^2]$ with $i \neq j$: there are $3\binom{n}{2}$ terms from the multinomial expansion, each equal to σ^4 by independence.

All other terms contain an odd power of some Y_i, which vanishes because $\mathbb{E}[Y_i] = 0$ and independence gives $\mathbb{E}[Y_i^k Y_j^\ell] = \mathbb{E}[Y_i^k]\mathbb{E}[Y_j^\ell] = 0$ when k is odd. Therefore

$$\mathbb{E}[S_n^4] = n\,\mathbb{E}[Y_1^4] + 3n(n-1)\sigma^4 \leq Cn^2, \tag{14}$$

where $C = \mathbb{E}[Y_1^4] + 3\sigma^4$.

Step 2: Summability via Markov. By Markov's inequality applied to $(\bar{X}_n - \mu)^4 = S_n^4/n^4$:

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\mathbb{E}[S_n^4]}{n^4\varepsilon^4} \leq \frac{C}{n^2\varepsilon^4}. \tag{15}$$

Since $\sum_{n=1}^{\infty} 1/n^2 < \infty$, we have $\sum_{n=1}^{\infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) < \infty$ for each $\varepsilon > 0$.

Step 3: Apply Borel–Cantelli. By the sufficient condition for almost sure convergence (Lesson 10, Corollary 5.2), $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$. □

Remark 4.3. The fourth moment proof above requires $\mathbb{E}[X_1^4] < \infty$, which is stronger than necessary. Kolmogorov's original proof requires only $\mathbb{E}[|X_1|] < \infty$ and proceeds by a truncation argument: replace X_i by $\tilde{X}_i = X_i \mathbf{1}_{|X_i| \le i}$, show the truncated and original sums agree eventually (using the first Borel–Cantelli lemma), then apply Kolmogorov's inequality to the truncated sum. See Durrett [2019], Theorem 2.4.1 for the complete argument.

SLLN and Stochastic Approximation

The SLLN guarantees almost sure convergence of sample averages, which is the starting point for stochastic approximation (SA) theory. The Robbins–Monro algorithm—the foundation of TD learning and Q-learning—can be viewed as computing a weighted, online version of sample averages. The SA convergence theorem (Lesson 14) extends the SLLN to settings with diminishing step sizes, Markov noise, and function approximation.

5 Central Limit Theorem

Central Limit Theorem

Theorem 5.1 (Central Limit Theorem). *Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_1] = \mu$ and $0 < \mathrm{Var}(X_1) = \sigma^2 < \infty$. Then*

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1). \tag{16}$$

Proof. Let $Y_i = (X_i - \mu)/\sigma$, so $\mathbb{E}[Y_i] = 0$ and $\mathbb{E}[Y_i^2] = 1$. Let $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$. We prove $Z_n \xrightarrow{d} \mathcal{N}(0,1)$ using characteristic functions.

Step 1: Characteristic function of Z_n. The characteristic function of Y_i is $\varphi(t) = \mathbb{E}[e^{itY_1}]$, which exists for all $t \in \mathbb{R}$. By independence:

$$\varphi_{Z_n}(t) = \mathbb{E}\left[e^{itZ_n}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{itY_i/\sqrt{n}}\right] = \left[\varphi\left(\frac{t}{\sqrt{n}}\right)\right]^n. \tag{17}$$

Step 2: Taylor expansion. Since $\mathbb{E}[Y_1] = 0$ and $\mathbb{E}[Y_1^2] = 1$, Taylor's theorem gives

$$\varphi(u) = 1 + iu\,\mathbb{E}[Y_1] - \frac{u^2}{2}\mathbb{E}[Y_1^2] + o(u^2) = 1 - \frac{u^2}{2} + o(u^2) \tag{18}$$

as $u \to 0$. Substituting $u = t/\sqrt{n}$:

$$\varphi\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right). \tag{19}$$

Step 3: Limit. Using the standard fact that $(1 + a_n/n)^n \to e^a$ when $a_n \to a$:

$$\varphi_{Z_n}(t) = \left[1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right]^n \to e^{-t^2/2} \quad \text{as } n \to \infty. \tag{20}$$

The function $e^{-t^2/2}$ is the characteristic function of $\mathcal{N}(0,1)$.

Step 4: Levy's continuity theorem. Since $\varphi_{Z_n}(t) \to e^{-t^2/2}$ pointwise for all $t \in \mathbb{R}$, and $e^{-t^2/2}$ is continuous at $t = 0$, Levy's continuity theorem guarantees $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. □

Remark 5.2 (Rate Interpretation). The CLT says $\bar{X}_n - \mu \approx \frac{\sigma}{\sqrt{n}} Z$ where $Z \sim \mathcal{N}(0,1)$. The estimation error is $O(1/\sqrt{n})$, and for a $(1-\alpha)$-confidence interval:

$$\bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \tag{21}$$

where $z_{\alpha/2}$ is the standard normal quantile.

CLT for Policy Evaluation

When using Monte Carlo to estimate $V^\pi(s)$ from n independent returns $G_1, \ldots, G_n$, the CLT gives

$$\sqrt{n} \frac{\bar{G}_n - V^\pi(s)}{\sigma_G} \xrightarrow{d} \mathcal{N}(0,1), \tag{22}$$

where $\sigma_G^2 = \mathrm{Var}(G_1)$. This provides approximate $(1-\alpha)$-confidence intervals: $\bar{G}_n \pm z_{\alpha/2}\, \hat{\sigma}_G/\sqrt{n}$. The CLT also quantifies the efficiency gain of variance reduction techniques (baselines, control variates): reducing σ_G^2 directly shrinks the confidence interval width.

6 Exercises

Exercise 6.1 ((*)). Let $a_n \geq 0$ for all n. Use the Monotone Convergence Theorem with the counting measure on $\mathbb{N}$ to prove that $\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} \sum_{n=1}^{N} a_n$. (This shows that MCT generalizes the elementary fact that partial sums of a non-negative series converge to the series sum.)

Exercise 6.2 ((*)). Let $\{f_n\}$ be a sequence of measurable functions on a probability space with $|f_n| \leq M$ a.s. for some constant M and all n. If $f_n \xrightarrow{\text{a.s.}} f$, prove that $\mathbb{E}[f_n] \to \mathbb{E}[f]$ using the DCT. What is the dominating function?

Exercise 6.3 ((**)). Prove the *reverse Fatou lemma*: if $X_n \leq Y$ a.s. with $\mathbb{E}[Y] < \infty$, then $\limsup_n \mathbb{E}[X_n] \leq \mathbb{E}[\limsup_n X_n]$.
Hint: Apply Fatou's lemma to $Y - X_n$.

Exercise 6.4 ((**)). Extend the WLLN to pairwise uncorrelated random variables: if $X_1, X_2, \ldots$ satisfy $\mathbb{E}[X_i] = \mu$, $\mathrm{Var}(X_i) \leq C$ for all i, and $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$, show that $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Exercise 6.5 ((**)). A policy π generates i.i.d. returns from state s with unknown mean $V^\pi(s)$ and variance $\sigma^2 = 4$. Using the CLT, determine the minimum number n of episodes needed so that the 95% confidence interval for $V^\pi(s)$ has half-width at most 0.1. (Use $z_{0.025} \approx 1.96$.)

Exercise 6.6 ((***)). Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = 0$. Define $\tilde{X}_i = X_i \mathbf{1}_{|X_i| \leq i}$.

(a) Use the first Borel–Cantelli lemma to show that $\mathbb{P}(X_i \neq \tilde{X}_i \text{ i.o.}) = 0$. *Hint:* Show $\sum_i \mathbb{P}(|X_i| > i) \leq \mathbb{E}[|X_1|] < \infty$.

(b) Conclude that $\frac{1}{n}\sum_{i=1}^n X_i$ and $\frac{1}{n}\sum_{i=1}^n \tilde{X}_i$ have the same almost sure limit (if either exists).

Exercise 6.7 ((***)). Consider a discounted MDP with $|R_t| \leq R_{\max}$ and $\gamma \in (0,1)$. Let $V_n(s) = \mathbb{E}[\sum_{t=0}^{n-1} \gamma^t R_{t+1} \mid S_0 = s]$ be the n-step truncated value function. Use the DCT to prove:

(a) $\lim_n V_n(s) = V^\pi(s)$ for each s.

(b) $|V^\pi(s) - V_n(s)| \leq \frac{\gamma^n R_{\max}}{1-\gamma}$ for each s.

7 Summary

- **MCT:** For non-negative increasing sequences, $\mathbb{E}[\lim X_n] = \lim \mathbb{E}[X_n]$.
- **Fatou:** For non-negative sequences, $\mathbb{E}[\liminf X_n] \leq \liminf \mathbb{E}[X_n]$. The inequality can be strict.
- **DCT:** Under a.s. convergence and domination by an integrable function, $\lim \mathbb{E}[X_n] = \mathbb{E}[\lim X_n]$ and $\mathbb{E}[|X_n - X|] \to 0$.
- **WLLN:** $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ for i.i.d. random variables with finite variance.
- **SLLN:** $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ for i.i.d. random variables with finite first moment.
- **CLT:** $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0,1)$ for i.i.d. random variables with finite variance. Error rate is $O(1/\sqrt{n})$.
- **RL connections:** WLLN justifies MC estimation; SLLN underpins SA convergence; CLT gives confidence intervals; DCT validates infinite-horizon value function definitions.

8 Further Reading

- **Lesson 10** establishes the modes of convergence and Borel–Cantelli lemmas used in the SLLN proof.
- **Lesson 12** on Markov chains uses the SLLN (via the ergodic theorem) for time-average convergence along trajectories.
- **Lesson 13** on martingale convergence extends the SLLN to dependent sequences via the martingale SLLN.
- Durrett [2019], Sections 2.2–2.4 gives the complete SLLN under the minimal first-moment hypothesis via truncation.
- Billingsley [1995], Sections 21–22 provides a thorough treatment of MCT, Fatou, and DCT with many variants and counterexamples.
- Folland [1999], Section 2.4 gives an elegant presentation of the convergence theorems for general measure spaces.
- Shiryaev [2016], Chapter III covers the CLT with extensions to the Lindeberg–Feller theorem for non-identically distributed summands.

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